

Exponential functions (2)

Exercise 1 :

Let f be the function that is defined, on $[0; +\infty[$, by $f(x) = (x + 1)e^{-x}$ and designate by (C) its representative curve in an orthonormal system $(O; \vec{i}, \vec{j})$. (unit: 2cm)

- 1- Calculate $\lim_{x \rightarrow +\infty} f(x)$ and deduce an asymptote of (C).
- 2- a) Calculate $f'(x)$ and set up the table of variations of f .
b) Calculate $f'(0)$ and interpret the result graphically.
- 3- a) Prove that the curve (C) has a point of inflection $W(1, \frac{2}{e})$.
b) Write an equation of the line (d) that is tangent to (C) at the point W.
- 4- Draw (d) and (C).
- 5- a) Calculate the real numbers a and b so that the function F defined on $[0; +\infty[$ by $F(x) = (ax + b)e^{-x}$ is an antiderivative of f .
b) Calculate, in cm^2 , the area of the region bounded by (C), the axis of abscissas and the two lines of equations $x = 0$ and $x = 1$.
- 6- Let g be the inverse function of f and designate by (G) the representative curve of g .
a) Draw (G) in the preceding system.
b) Find an equation of the tangent to the curve (G) at the point of abscissa $\frac{2}{e}$.

Exercise 2 :

Consider the function f defined on $\mathbb{R}$ as $f(x) = 3 - \frac{4}{e^{2x} + 1}$.

Let (C) be its representative curve in an orthonormal system (unit 2 cm).

- 1) Calculate $\lim_{x \rightarrow -\infty} f(x)$, $\lim_{x \rightarrow +\infty} f(x)$ and deduce the asymptotes to (C).
- 2) Prove that f is strictly increasing over $\mathbb{R}$ and set up its table of variations.
- 3) The curve (C) has a point of inflection W with abscissa 0. Write an equation of (T), the tangent to (C) at the point W.
- 4) a- Calculate the abscissa of the point of intersection of (C) with the x-axis.
b- Draw (T) and (C).
- 5) a- Verify that $f(x) = -1 + \frac{4e^{2x}}{e^{2x} + 1}$ and deduce an antiderivative F of f .
b- Calculate, in cm^2 , the area of the region bounded by the curve (C), the x- axis, the y-axis and the line with equation $x = \ln 2$.
- 6) The function f has over $\mathbb{R}$ an inverse function g . Denote by (G) the representative curve of g .
 - a- Specify the domain of definition of g .
 - b- Show that (G) has a point of inflection J whose coordinates to be determined.
 - c- Draw (G) in the same system as (C).
 - d- Determine $g(x)$ in terms of x .

Exercise 3 :

Let f be the function defined, on $] -\infty ; +\infty[$, by $f(x) = x + 2 - \frac{3}{1 + e^x}$.

(C) is the representative curve of f in an orthonormal system $(O; \vec{i}, \vec{j})$.

- 1) a- Calculate $\lim_{x \rightarrow -\infty} f(x)$; Show that the line (d_1) with equation $y = x - 1$ is an asymptote to (C) and specify the position of (d_1) relative to (C).
b- Calculate $\lim_{x \rightarrow +\infty} f(x)$; Show that the line (d_2) with equation $y = x + 2$ is an asymptote to (C) and specify the position of (d_2) relative to (C).
- 2) Prove that the point $I(0; \frac{1}{2})$ is a center of symmetry of (C).
- 3) Show that f is strictly increasing on $] -\infty ; +\infty[$ and set up its table of variations.
- 4) Draw (d_1) , (d_2) and (C).
- 5) a - Verify that $f(x) = x + 2 - \frac{3e^{-x}}{1 + e^{-x}}$.
b - Calculate the area $A(\lambda)$ of the region bounded by the curve (C), the asymptote (d_2) and the two lines with equations $x = 0$ and $x = \lambda$, where $\lambda > 0$, then calculate $\lim_{\lambda \rightarrow +\infty} A(\lambda)$.
- 6) Designate by g the inverse function of f on $] -\infty ; +\infty[$; (G) is the representative curve of g .
a- Verify that $E(1 + \ln 2; \ln 2)$ is a point on (G).
b- Calculate the slope of the tangent to (G) at E.

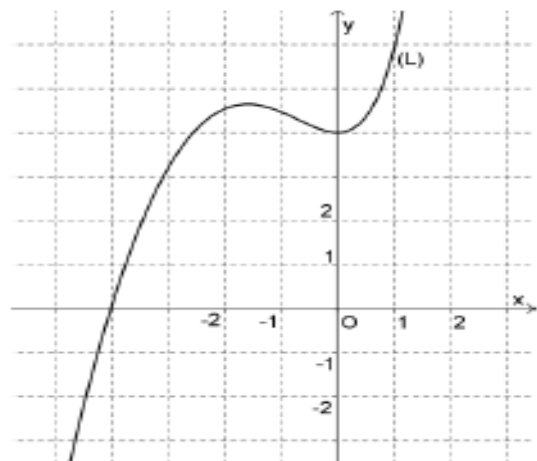
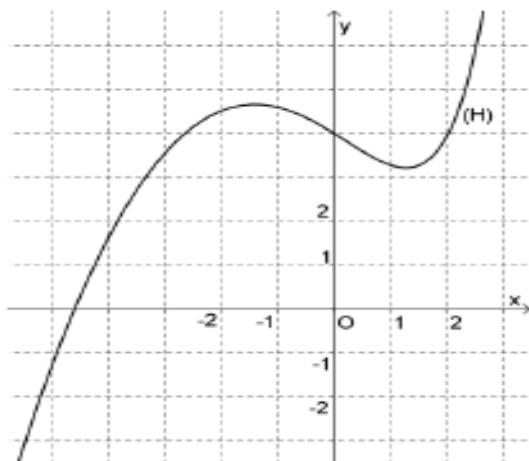
Exercise 4

Let f be the function defined on $\mathbb{R}$ as: $f(x) = x + 2 - 2e^x$. Denote by (C) its representative curve in an orthonormal system $(O; \vec{i}, \vec{j})$.

- 1) a- Determine $\lim_{x \rightarrow -\infty} f(x)$.
b- Show that the line (D) with equation $y = x + 2$ is an asymptote to (C).
c- For all x in $\mathbb{R}$, show that the curve (C) is below the line (D).
- 2) Determine $\lim_{x \rightarrow +\infty} f(x)$ and calculate $f(1.5)$.
- 3) Calculate $f'(x)$ and set up the table of variations of f .
- 4) Show that the equation $f(x) = 0$ has, in $\mathbb{R}$, exactly two roots 0 and α .
Verify that $-1.6 < \alpha < -1.5$.
- 5) Draw (D) and (C).
- 6) Denote by $A(\alpha)$ the area of the region bounded by (C) and the x-axis.

Show that $A(\alpha) = \left(-\frac{\alpha^2}{2} - \alpha \right)$ square units.

- 7) Let g be a function defined on $\mathbb{R}$ with: $g'(x) = -2f(x)$.
One of the two curves (H) and (L) given below represents the function g .
Choose it with justification.



Exercise 5 :

Consider the function f defined over $\mathbb{R}$ by $f(x) = (x+1)^2 e^{-x}$ and denote by (C) its representative curve in an orthonormal system $(O; \vec{i}, \vec{j})$.

- 1) a- Determine $\lim_{x \rightarrow -\infty} f(x)$ and calculate $f(-2)$.
b- Determine $\lim_{x \rightarrow +\infty} f(x)$ and deduce an asymptote to (C) .
- 2) Show that $f'(x) = (1-x^2)e^{-x}$ and set up the table of variations of f .
- 3) The line (d) with equation $y = x$ intersects (C) at a point with abscissa α .

Verify that $1.4 < \alpha < 1.5$.

- 4) Draw (d) and (C) .
- 5) Let F be the function defined on $\mathbb{R}$ by $F(x) = (px^2 + qx + r)e^{-x}$.
a- Calculate p , q and r so that F is an antiderivative of f .
b- Calculate the area of the region bounded by (C) , the axis of abscissas and the two lines with equations $x = 0$ and $x = 1$.
- 6) The function f has over $[1; +\infty[$ an inverse function h . Determine the domain of definition of h and draw its representative curve in the same system as (C) .

Exercise 6:

Consider the function f defined on $\mathbb{R}$ by $f(x) = \frac{e^x}{e^x + 1}$. Designate by (C) the representative curve of f in an orthonormal system $(O; \vec{i}, \vec{j})$.

- 1) Calculate $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow +\infty} f(x)$. Deduce the asymptotes of the curve (C) .
- 2) Calculate $f'(x)$ and set up the table of variations of f .
- 3) Show that $f''(x) = \frac{e^x(1-e^x)}{(1+e^x)^3}$. Prove that (C) has a point of inflection I to be determined.
- 4) Write an equation of the tangent (T) to (C) at the point I .
- 5) Draw (T) and (C) .
- 6) The function f has on $\mathbb{R}$ an inverse function g .
a- Draw the representative curve (G) of g in the given system.
b- Verify that $g(x) = \ln\left(\frac{x}{1-x}\right)$.
c- (G) and (C) intersect at a point with abscissa α . Calculate, in terms of α , the area of the region bounded by (C) , (G) and the two axes of coordinates.

Exercise 7:

Consider the function f defined over $\mathbb{R}$ as $f(x) = (1 - x)e^x + 2$.

Denote by (C) its representative curve in an orthonormal system $(O; \vec{i}; \vec{j})$.

1) a) Determine $\lim_{x \rightarrow -\infty} f(x)$. Deduce an asymptote (d) to (C).

b) Determine $\lim_{x \rightarrow +\infty} f(x)$, then calculate $f(1)$ and $f(2)$.

2) a) Verify that $f'(x) = -xe^x$ and set up the table of variations of the function f .

b) Prove that the curve (C) has an inflection point I whose coordinates should be determined.

3) Draw (d) and (C).

4) Denote by (Δ) the line with equation $y = 2x$.

a) Verify that $f(x) - 2x = (e^x + 2)(1 - x)$. Study, according to the values of x , the relative positions of (C) and (Δ).

b) Find an antiderivative F of the function f .

c) Draw (Δ), then calculate the area of the region bounded by the curve (C), the y-axis and the line (Δ)

5) Let g be the function given as $g(x) = \ln[f(x) - 2]$.

Denote by (G) the representative curve of g in the system $(O; \vec{i}; \vec{j})$.

a) Verify that the domain of definition of g is $] -\infty; 1[$.

b) Is there a point on (G) where the tangent to (G) is parallel to the line (Δ)? Justify.